

Bucket Sort / Count sort

Given n integers in range $\{0, 1, \dots, m-1\}$
we can sort in $O(n+m)$ time
by scanning the members and then scanning
the buckets in order.

Best case when $n \sim m$

① 3 8 4
 ② 9 1 6
 ③ 5 2 9
 ④ 1 8 4
 ⑤ 2 6 8
 ⑥ 7 7 5
 ⑦ 3 0 5
 ⑧ 4 1 5
 ⑨ 8 2 9
 ⑩ 6 7 0
 2 1 0

⑩ 6 7 0
 ① 3 8 4
 ④ 1 8 4
 ⑥ 7 7 5
 ⑦ 3 0 5
 ⑧ 4 1 5
 ② 9 1 6
 ⑤ 2 6 8
 ③ 5 2 9
 ⑨ 8 2 9

⑦ 3 0 5
 ⑧ 4 1 5
 ② 9 1 6
 ③ 5 2 9
 ⑨ 8 2 9
 ⑤ 2 6 8
 ⑩ 6 7 0
 ⑥ 7 7 5
 ① 3 8 4
 ④ 1 8 4

④ 1 8 4
 ⑤ 2 6 8
 ⑦ 3 0 5
 ① 3 8 4
 ⑧ 4 1 5
 ③ 5 2 9
 ⑩ 6 7 0
 ⑥ 7 7 5
 ⑨ 8 2 9
 ② 9 1 6

Stable sorting: Numbers having same values are preserved as per their original order

① 1 0 1 1 0 0 1 0
 ② 0 1 0 1 1 0 1 0
 ③ 1 0 0 1 0 1 0 1
 ④ 0 0 0 0 1 1 1 0
 ⑤ 0 1 0 1 0 1 0 1

① 0
 ② 0
 ④ 0
 ③ 1
 ⑤ 1

③ 0 1
 ⑤ 0 1
 ① 1 0
 ② 1 0
 ④ 1 0

① 0 1 0
 ② 0 1 0
 ③ 1 0 1
 ⑤ 1 0 1
 ④ 1 1 0

⋮

⋮

Hex Notation

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
0 1 2 3 4 5 6 7 8 9 A B C D E F

① B 2
② 5 A
③ 9 5
④ 0 E
⑤ 5 5

B 2
9 5
5 5
5 A
0 E

④ 0 E
⑤ 5 5
② 5 A
③ 9 5
① B 2

By choosing $\text{radix} = \log_{16} n$, i.e. n
we can sort n numbers in $[0, \dots, n^k - 1]$
in time $O(k \cdot n)$

Given a set of strings over some
alphabet, say English, how would

you order them

Eg. you, order, them,
the
order
them
the
you

Lexicographic ordering can be achieved
using radix sort if we introduce
a character and pad up all strings
to be of the same length

Suppose we have n strings

s_1 s_2 s_3 $\dots$ s_n having lengths
 l_1 l_2 l_3 $\dots$ l_n

$$-1 \quad v_2 \quad v_3 \quad \dots \quad v_n \quad l_i \leq L$$

$$\sum l_i = N$$

How would you compare
 $n \times L : N \leftarrow$ input size
 by padding

Graphs
 V : vertices
 E : edges

Directed
 Undirected



$$|V| = n$$

$$|E| = m$$



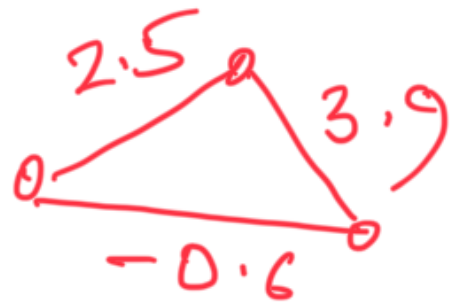
$$E = \{ (u_i, v_i) \}$$

$u_i, v_i \in V$

for directed
 $u_i \rightarrow v_i$

DFS

BFS



Weighted $W: E \rightarrow \mathbb{R}$

Unweighted

Given an undirected graph $G = (V, E)$

what is the notion of connectivity?

$\forall u, w \in V$ $u \rightsquigarrow w$ and $w \rightsquigarrow u$
undirected

How do we...

... know if the given graph
is connected?

DFS / BFS

Time $O(|V| + |E|)$

using adjacency list $= O(n + m)$

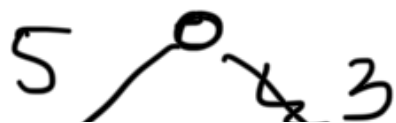
What is a spanning Tree of G

"Connected" Subgraph with minimum
edges
 $\Rightarrow$ Tree

Given a weighted graph $G = (V, E, w)$

find the spanning with least cost

(sum of weights of edges)



~~Summer~~

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